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AI-Powered Study Assistant

Final Project Report

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Abstract

Artificial Intelligence has become increasingly important in educational technologies because it can automate learning support and improve student engagement. This project presents an AI-powered study assistant that transforms raw study notes into structured educational content. The system generates three main outputs: a concise summary, a quiz, and a visual mind map image.

The application uses **Claude Haiku** as the main Large Language Model for text-based tasks, including summarization and quiz generation. It also uses Pollinations.ai as an image-generation service to create educational mind maps. The project applies prompt engineering and chained prompting to control the format and quality of the generated outputs.

The system demonstrates how generative AI can support students by organizing information, testing understanding, and visualizing concepts. Experimental testing showed that the application can generate useful summaries, relevant quiz questions, and visual mind maps for different study topics. Although the system has limitations related to image accuracy, API dependency, and lack of persistent storage, it shows strong potential as a lightweight AI-assisted learning tool.

Chapter 1

Introduction

Students often need to review large amounts of study material in a limited time. Manually summarizing notes, creating practice questions, and building visual diagrams can be time-consuming. This creates a need for intelligent study tools that can help students organize and understand educational content more efficiently.

This project introduces an AI powered study assistant that accepts study notes from the user and automatically generates three types of outputs: a bullet-point summary, a quiz, and a visual mind map. The main purpose of the system is to support learning by combining text-based understanding with visual representation.

The problem addressed by this project is that students often lack fast and structured tools for transforming unorganized notes into useful study materials. Traditional note revision depends heavily on manual effort. Generative AI can reduce this effort by producing organized and personalized learning content.

The objectives of this project are:

- To generate concise summaries from user-provided study notes.
- To create quizzes that help students test their understanding.
- To generate visual mind maps that support concept visualization.
- To use more than one AI modality, including text and image generation.
- To demonstrate prompt engineering as a core generative AI technique.

The project uses **Claude Haiku** for text generation and **Pollinations.ai** for image generation. Instead of training a new model, the system depends on carefully designed prompts to guide model behavior.

Chapter 2

Related Work

Generative AI has recently become an important area in educational technology. Large Language Models have been used for summarization, tutoring, feedback generation, and quiz creation. Recent research shows that LLMs can support educational tasks by reducing teacher workload and providing students with personalized learning materials.

Maity et al. (2025) investigated whether LLMs can generate school-level educational questions from textbook content. Their work supports the idea that LLMs can be used to automate question generation in education. Similarly, Mucciaccia et al. (2025) explored automatic multiple-choice question generation and validation using LLMs and prompt engineering techniques.

Biancini et al. (2025) studied multiple-choice question generation using large language models and reported that AI-generated questions can be useful when guided by proper prompts. Fu (2025) introduced ConQuer, a framework for concept-based quiz generation, which shows the importance of generating questions based on key learning concepts rather than random text extraction.

Multimodal generative AI has also become relevant in education. Recent studies have examined the use of multimodal AI systems that combine text, image, and other data formats to support learning. For example, research on multimodal generative AI in education shows that combining modalities can create richer learning experiences.

In addition, diffusion-based image generation has been widely studied for visual content creation. While diffusion models can generate useful educational visuals, they may still struggle with precise text rendering inside images. This is relevant to the current project because the mind map output is generated as an image and may contain imperfect labels.

Compared with these related works, this project combines summarization, quiz generation, and mind map generation in one simple pipeline. It is designed as a lightweight study assistant that uses accessible APIs and prompt engineering rather than model training.

Chapter 3

Methodology

3.1 System Overview

The AI-powered study assistant was implemented as a Python notebook. The system follows a pipeline structure where user input is processed through multiple functions. Each function performs a specific task, and the final output is shown to the user in a readable format.

The main system workflow is:

1. The user enters a topic name.
2. The user pastes study notes.
3. The system sends the notes to Claude Haiku for summarization.
4. The system sends the same notes to Claude Haiku again for quiz generation.
5. The system sends the topic to Pollinations.ai for mind map image generation.
6. The system displays the generated summary, quiz, and mind map.

3.2 AI Approach

The project uses a multimodal generative AI approach. It includes two modalities:

- **Text modality:** Claude Haiku is used for summary and quiz generation.
- **Image modality:** Pollinations.ai is used to generate visual mind map images.

The main generative AI technique used in the project is prompt engineering. No model training or fine-tuning is performed. Instead, the system uses carefully written prompts to guide the output format.

3.3 Model Design

The model design is based on separate functional components. Each component is responsible for one part of the system.

3.3.1 Library Installation

The first cell installs the required Python libraries:

- `anthropic`
- `requests`
- `pillow`

The `anthropic` library is used to connect with Claude.
The `requests` library is used to send requests to Pollinations.ai.
The `pillow` library is used to open and display generated images.

3.3.2 Authentication

The system creates an Anthropic client using an API key. This client allows the notebook to communicate with Claude Haiku. The API key works as an authentication method that identifies the user to Anthropic servers.

3.3.3 Summarization Function

The `summarize()` function receives study notes as input. It sends the notes to Claude Haiku with a prompt that asks the model to act as a study assistant. The required output is a concise summary with bullet points and one important concept sentence.

3.3.4 Quiz Generation Function

The `generate_quiz()` function sends the same study notes to Claude Haiku with a different prompt. This prompt asks the model to generate three multiple-choice questions and two short-answer questions. Each multiple-choice question includes four options and the correct answer.

This demonstrates chained prompting because the same input is used with different prompts to generate different outputs.

3.3.5 Mind Map Image Function

The `generate_mindmap_image()` function sends the topic name to Pollinations.ai. The function creates a descriptive prompt such as an educational mind map with a clean background and colorful branches. Pollinations.ai then returns a generated image, which is displayed in the notebook.

3.3.6 Main Pipeline

The `study_assistant()` function connects all previous functions. It calls the summarization function, quiz generation function, and mind map generation function in sequence. This makes the system easier to run and understand.

Chapter 4

Experimental Results and Discussion

4.1 Testing Setup

The system was tested using different educational topics such as Photosynthesis, the French Revolution, and Light. For each topic, study notes were provided as input. The system then generated a summary, quiz, and mind map.

4.2 Evaluation Metrics

The project was evaluated using qualitative metrics because the system focuses on educational usefulness rather than numerical prediction accuracy. The evaluation criteria included:

- **Relevance:** Whether the generated content matched the input notes.
- **Clarity:** Whether the output was easy to read and understand.
- **Structure:** Whether the summary and quiz followed the required format.
- **Usefulness:** Whether the generated output could help students study.
- **Visual quality:** Whether the mind map image was understandable.

4.3 Results Analysis

The summary generation results were generally clear and useful. Claude Haiku was able to identify the main ideas from the notes and present them in organized bullet points. This output is helpful for quick review before exams.

The quiz generation results were also effective. The system generated multiple-choice questions and short-answer questions that were related to the provided notes. This supports active recall, which is important for improving learning.

The mind map generation results were visually useful, but less precise than the text outputs. The generated images helped represent the topic visually, but some labels could be blurry or inaccurate. This is a known limitation of many image-generation models when producing text inside images.

4.4 Position Relative to Related Work

Compared with related studies on automatic quiz generation, this project uses a simpler implementation but provides multiple outputs in one workflow. It does not only generate questions; it also produces summaries and visual mind maps. Compared with larger multimodal educational systems, this project is lightweight and easier to run in a notebook environment.

The project confirms that prompt engineering can be effective for educational content generation. However, it also shows that text-generation models are currently more reliable than image-generation models for precise academic content.

Chapter 5

Conclusion

This project successfully developed an AI-powered study assistant that generates summaries, quizzes, and visual mind maps from user-provided study notes. The system uses Claude Haiku for text generation and Pollinations.ai for image generation.

The project demonstrates the value of prompt engineering and chained prompting in educational applications. The results show that generative AI can help students organize information, review key concepts, and test their understanding.

Overall, the project achieved its main objectives and provides a practical example of using multimodal generative AI to support learning.

Chapter 6

Limitations and Challenges

The project has several limitations:

- The generated mind map images may contain blurry or inaccurate text.
- The quality of the output depends on the quality of the input notes.
- The current version supports English only.
- The system does not save outputs automatically between sessions.
- The application depends on external APIs, so availability and speed may vary.
- API keys must be handled carefully to protect user privacy and security.

Chapter 7

Future Work

Future improvements can make the system more useful and complete. Possible improvements include:

- Adding Arabic language support.
- Allowing users to upload PDF, Word, or PowerPoint files directly.
- Saving generated summaries, quizzes, and images automatically.
- Improving mind map accuracy using structured diagram generation.
- Adding difficulty levels for quizzes.
- Adding progress tracking for students.
- Creating a web interface instead of using only a notebook.
- Adding answer explanations for quiz questions.

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